

LIM ZHAO WEI KENNETH

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PROFESSIONAL SUMMARY

Results-driven AI Engineer with a proven track record of architecting and operationalizing production-scale AI/ML systems within regulated enterprise environments. Specializing in high-performance backend microservices, inference optimization, and semantic retrieval architectures using transformer-based models. Experienced in MLOps, integrating model observability and automated CI/CD pipelines to bridge the gap between R&D and production, ensuring robust, enterprise-grade production deployments.

TECHNICAL SKILLS

AI/ML : Transformers (BERT), LLM Foundations, Semantic Search, Embedding Pipelines, RAG, NLP, Reranking, NER, Prompt Engineering, Vector Databases

Backend & API Development: Python, SQL, FastAPI, RESTful APIs, Microservices, JavaScript, Bash, Langchain, LLM APIs

MLOps & Cloud: AWS(Sagemaker, S3, Redshift, Glue), Docker, MLflow, CI/CD pipelines

Monitoring & Data Engineering: Model Drift Detection, ETL pipelines, Tableau, Streamlit

Development Practices: Unit & Integration testing, Agile Methodologies, Git, Gitlab, Jira, Confluence

WORK EXPERIENCE

SkillsFuture Singapore – AI/ML Engineer

Jan 2025 – Present

- Architected a production-scale Semantic Retrieval pipeline on **AWS Sagemaker and S3** to automate transformer-based text embeddings generations for national labour market analytics
- Operationalised Advanced Retrieval Logic by translating **RAG**-based proof-of-concepts into production-ready framework, reducing manual tagging effort by **35%** and turnaround time by **2 weeks**.
- Established enterprise **MLOps SLAs**, implementing **CI/CD pipelines**, pre-production testing and controlled rollouts to reduce deployment-related issues by **25%**
- Engineered high-performance backend services via **Swagger REST API** endpoints, achieving **P95 inference latency of ~1s** through optimized data handling and efficient prediction-log compression files
- Led the design of model monitoring and observability for fraud detection system to enable near-real-time telemetry, performance alerting, **drift detection**, and incident triage via Tableau dashboards
- Streamlined an **ETL Knowledge Graph pipeline** by transitioning experimental logic to production-ready framework through refined **LLM prompt integration** from backend APIs to ensure consistent entity extraction for cross-sector skills mapping

AI Singapore – AI Engineer Apprentice

Jan 2024 – Oct 2024

- Developed an **end-to-end ML pipeline** to automate treatment classification and diagnosis prediction, enhancing clinical workflow efficiency.
- Fine-tuned a **biomedical BERT Models** for **Named Entity Recognition(NER)** to extract infection diagnoses from medical reports, improving feature F1 score by **25%**.
- Co-developed a boosting classification model integrating **NER** predictions and clinical features to enhance efficiency and responsiveness for medical intervention planning
- Containerized ML pipeline using **Docker, MLflow and FastAPI**, to enable reproducible and scalable deployments

PROJECTS

Travel Itinerary Generator

- Built an AI-powered **multi-agent pipeline** using **CrewAI** and proprietary **LLMs** with robust Prompt Engineering to orchestrate specialized agents into contextual travel itineraries, wrapped in prototype frontend interface

EDUCATION & CERTIFICATIONS

Georgia Institute of Technology: Online M.S. in Computer Science.

Jan 2026 - Present

Nanyang Technological University: B.Eng. in Materials Engineering (Honours - Distinction)

Certifications: Chartered Assoc. AI Engineer (AI Singapore), GA Data Science Immersive